

Everything Summer '26

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1 Bunch of Definitions

Definition 1.1 (Group). A group $(G, \cdot)$ is a set G with a binary operation $\cdot : G \times G \rightarrow G$ with a multiplicative identity and an inverse for all $g \in G$, and the binary operation is associative.

Definition 1.2 (Vector Space). A vector space over a field F is a non-empty set V together with a binary operation and a binary function that satisfy the following eight axioms:

- (1-4). associativity, commutativity, identity element, inverse elements of vector addition;
- (5). compatibility of scalar multiplication with field multiplication;
- (6). identity element of scalar multiplication;
- (7). distributivity of scalar multiplication with respect to vector addition;
- (8). distributivity of scalar multiplication with respect to field addition.

Definition 1.3 (Ring). A ring $(R, +, \cdot)$ is a set R with two binary operations called addition and multiplication such that $(R, +)$ is an abelian group, the multiplication $\cdot$ is associative, and the left and right distributive properties hold.

We usually care about *unital rings*: there exists a multiplicative identity $1 \in R$ with $1 \cdot r = r \cdot 1 = r$ for all $r \in R$.

Definition 1.4 (Algebra). An algebra over $\mathbb{C}$ is a vector space (over $\mathbb{C}$) that is also a ring.

Definition 1.5 (Group Algebra). The group algebra $\mathbb{C}[G]$ is the complex vector space with basis $|g\rangle$, $|g\rangle \in G$ and multiplication on basis elements induced by group multiplication $|g\rangle \cdot |h\rangle = |gh\rangle$.

Multiplication on all of $\mathbb{C}[G]$ comes from extending the multiplication to arbitrary linear combinations of basis elements. The multiplicative unit is $|e\rangle$, where e is the identity element of G .

Definition 1.6 (Dual of group algebra). *The dual of a group algebra $\mathbb{C}[G]$ is the algebra $\hat{\mathbb{C}}[G]$ of linear functions of a finite group: it is the complex vector space with indicator function basis $|I_g\rangle$, $g \in G$ with multiplication of basis elements given by $|I_g\rangle \cdot |I_h\rangle = \delta_{g,h}|I_g\rangle$, where δ is the Kronecker delta function.*

$$I_g : G \rightarrow \mathbb{C}$$

$$h \mapsto \begin{cases} 1 & h = g \\ 0 & h \neq g \end{cases}$$

The multiplicative unit is $\sum_g |I_g\rangle$.

Definition 1.7 (Doubled algebra). *We build a “doubled” algebra DG that has a copy of each of $\mathbb{C}[G]$ and $\hat{\mathbb{C}}[G]$. As a vector space, $DG = \hat{\mathbb{C}}[G] \otimes \mathbb{C}[G]$.*

$\mathbb{C}[G]$, its dual $\hat{\mathbb{C}}[G]$, and the doubled algebra DG are all Hopf Algebras (we skip the details for now).

Definition 1.8 (Fundamental Group). *The fundamental group $\pi_1(X, x)$ of a space X with respect to a basepoint x is the set of homotopy (continuous deformation) classes of loops based at x with multiplication inherited from concatenation of paths.*

2 Kitaev Quantum Double Model

The flow: Fix a lattice $\rightarrow$ edges with quantum degrees of freedom $\rightarrow$ local operators and Hamiltonian $\rightarrow$ ground state and excitations.

State space: Lattice with V, E, and F: the set of vertices, edges, and faces, respectively. Assign an orientation to each $e \in E$ (arbitrary).

$$\text{Sites } \mathcal{S} = \{(p, v) \mid p \in F, v \in \partial p\}.$$

At every edge $e \in E$, we put a $|G|$ -dimensional qudit $\mathcal{H}_e = \mathbb{C}[G]$ with computational basis $\{|g\rangle\}$. Total Hilbert space $\mathcal{H}_{\text{tot}} = \otimes_{e \in E} \mathcal{H}_e$ with $\dim(\mathcal{H}_{\text{tot}}) = |G|^{|E|}$. Also,

$$|g_{e_1} g_{e_2} \dots g_{e_{|E|}}\rangle := |g_{e_1}\rangle \otimes |g_{e_1}\rangle \otimes \dots \otimes |g_{e_{|E|}}\rangle.$$

Note: The ring multiplication structure of $\mathbb{C}[G]$ comes into play in the local operators.

Hamiltonian: $A_g(v)$ is defined as follows: (Figure to be included)

$$|\dots g_1 g_2 g_3 g_4 \dots\rangle \mapsto |\dots g g_1 g g_2 g_3 g^{-1} g_4 g^{-1} \dots\rangle.$$

(Note that $A_{g_1} A_{g_2} = A_{g_1 g_2}$ since $g_2^{-1} g_1^{-1} = (g_1 g_2)^{-1}$.) We define the vertex operator $A(s)$ where $s \in \mathcal{S}$:

$$A(s) := A(v) = \frac{1}{|G|} \sum_g A_g(v)$$

This is a projector. *Proof:* $A^2(s) = \frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1} A_{g_2}(v) = \frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1 g_2}(v)$.

Now, due to the rearrangement of finite groups, $\frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1 g_2}(v) = \frac{1}{|G|^2} \sum_{g_1} \left(\sum_g A_g(v) \right) = \frac{1}{|G|^2} \cdot |G| \cdot \sum_g A_g(v) = A(v)$. $\square$

$B_h(v)$ is defined as follows: (Figure to be included)

If $h = e$, the relation $g^{o(e_1)} g^{o(e_2)} g^{o(e_3)} g^{o(e_4)} = e$ can be cyclically permuted ($o(e)$ is the orientation of edge e). Hence, the starting vertex for traversing the plaquette p **does not** matter.

$A_g(s)$ and $B_h(s)$ do not commute but satisfy

$$A_g(s)B_h(s) = B_{ghg^{-1}}(s)A_g(s). \quad (1)$$

In a more convenient notation, $A_g(v)B_h(p) = B_{ghg^{-1}}(p)A_g(v)$. This means that we start traversing the plaquette p counterclockwise from vertex v . (Figure to be included)

Firstly,

$$\begin{aligned} A_g(v)B_h(p)|\dots g_1 g_2 g_3 g_4 \dots\rangle &= \delta_{h,g_2^{-1}g_3g_4g_1^{-1}}A_g(v)|\dots g_1 g_2 g_3 g_4 \dots\rangle \\ &= \delta_{h,g_2^{-1}g_3g_4g_1^{-1}}|\dots gg_1 gg_2 g_3g^{-1} g_4g^{-1} \dots\rangle. \end{aligned}$$

Here, we ignore the effects of $A_g(v)$ on the rest of the vertices since they are irrelevant in this context. Secondly,

$$\begin{aligned} B_{ghg^{-1}}(p)A_g(v)|\dots g_1 g_2 g_3 g_4 \dots\rangle &= B_{ghg^{-1}}(p)|\dots gg_1 gg_2 g_3g^{-1} g_4g^{-1} \dots\rangle \\ &= \delta_{ghg^{-1},(g_2g^{-1})^{-1}g_3g_4(gg_1)^{-1}}A_g(v)|\dots gg_1 gg_2 g_3g^{-1} g_4g^{-1} \dots\rangle. \end{aligned}$$

Finally, we note that the conditions

$$\begin{aligned} ghg^{-1} &= (g_2g^{-1})^{-1}g_3g_4(gg_1)^{-1} \\ \Leftrightarrow ghg^{-1} &= g(g_2^{-1}g_3g_4g_1^{-1})g^{-1} \\ \Leftrightarrow h &= g_2^{-1}g_3g_4g_1^{-1} \end{aligned}$$

are all equivalent, meaning that

$$A_g(v)B_h(p) = B_{ghg^{-1}}(p)A_g(v) \quad \square$$

If we flip the direction of e_1 , we get $h = g_2^{-1}g_3g_4g_1$ and $ghg^{-1} = (g_2g^{-1})^{-1}g_3g_4(g_1g^{-1})$, which are also equivalent. The argument is similar if we flip e_2 .

They satisfy the same relations as the generators of DG, which implies that the algebra of local operators at a (!) site $s = (p, v)$ generated by A_g and B_h form an $N = |G|^{|E|}$ (the exponent $|E|$, for ‘a’ site, how?) dimensional matrix representation of the quantum double algebra DG given by the homomorphism

$$\begin{aligned} \rho_s : DG &\rightarrow GL(N, \mathbb{C}) \\ D_{e, g} &\mapsto A_g(v) \\ D_{h, e} &\mapsto B_h(p). \end{aligned}$$

Altogether, the quantum double **Hamiltonian** is given by

$$H = - \sum_{v \in V} A(v) - \sum_{p \in P} B(p), \quad (2)$$

where $B(p) := B_e(p)$. We can check from the non-commuting relation of A_g and B_h that all the projectors in the Hamiltonian commute with each other.